

Medium Access Control and Pure Aloha: Technical Analysis

Summary

The **Medium Access Control (MAC)** sublayer of the Data Link Layer (DLL) is responsible for managing **access to shared broadcast media**, preventing collisions, and ensuring efficient use of the communication channel. Among the random access protocols, **Pure Aloha**—developed in 1970 at the University of Hawaii—serves as a foundational model for **wireless LANs and shared medium networks**. Collisions are resolved using the **Binary Exponential Backoff (BEB)** algorithm, but this approach can lead to the **Capture Effect**, where one station dominates the channel.

1. Data Link Layer Architecture

The DLL is divided into two sublayers:

1.1 Logical Link Control (LLC)

- Positioned **between Network Layer and MAC**.
- Functions:
 - **Error Control:** Detection and correction (e.g., CRC).
 - **Flow Control:** Prevents sender from overwhelming the receiver (e.g., Stop-and-Wait, Go-Back-N, Selective Repeat).

1.2 Medium Access Control (MAC)

- Positioned **between LLC and Physical Layer**.
- Responsible for **managing access to a shared medium**, especially in broadcast networks.
- Core function: ensure **one station transmits at a time** to minimize collisions.

2. Fundamentals of Medium Access Control

Access Control ensures efficient sharing of communication channels:

- **Point-to-Point Links:** Dedicated connections, full-duplex prevents collisions.

- **Broadcast/Shared Links:** Multiple stations share the medium; simultaneous transmissions cause **collisions**.

Key Definition:

Medium Access Control = Method of regulating station transmissions to avoid data collisions on a shared medium.

GATE Examination Trends (30-Year Analysis):

Protocol	Questions Asked
Pure Aloha	1
CSMA / CSMA-CA / FDMA/CDMA	0
CSMA/CD	18
Polling	1
TDMA	2

3. Random Access Protocols

Random Access (Contention-Based) protocols allow stations to transmit **without centralized control**:

- **Equal Opportunity:** No station has priority.
- **Random Transmission:** Stations transmit whenever ready.
- **Collision Risk:** Simultaneous transmissions result in lost frames.
- **Procedure-Based:** Protocols define retransmission rules to reduce repeated collisions.

4. The Aloha Protocol

Developed in 1970, **Aloha** is foundational for shared medium networks.

4.1 Core Assumptions

- **Equal Frame Size:** All frames are identical to simplify throughput calculation.
- Two variants:
 - **Pure Aloha:** Transmit anytime.
 - **Slotted Aloha:** Transmit in synchronized time slots.

4.2 Pure Aloha Mechanics

1. **Transmit:** Station sends a frame immediately.
2. **Wait for ACK:** Station expects acknowledgment from receiver.
3. **Time-Out:** Maximum round-trip propagation delay ($2 \times \text{Propagation Delay}$).
4. **Retransmission:** If no ACK, retransmit frame after backoff.

5. Collision Resolution: Binary Exponential Backoff (BEB)

5.1 Backoff Calculation

- After collision, station chooses **random wait time**:

$$T_B = k \times \text{Slot Duration}$$

- **k:** Random integer in $[0, 2^n - 1]$, where **n = number of collisions** for that frame.
- **Max Retries:** Typically 15 attempts.

5.2 Probability of Success

Collision Number (n)	k Range	Success Probability (Winning Station)	Probability of Continued Collision
1	[0,1]	25%	50%
2	[0,3]	62.5%	25%
3	[0,7]	81.25%	12.5%

As collisions increase, **probability of successful transmission rises exponentially**.

6. Capture Effect

Definition: A single station can dominate the channel due to BEB probability mechanics.

- **Mechanism:**
 - Station A wins first contention ($n=1$), transmits successfully.
 - Its next frame resets to $n=1$.

- Competing station B has higher n due to previous collisions.
 - Station A now has a higher chance of winning, reinforcing its advantage.
- **Outcome:** Other stations experience **starvation**, while the successful station “captures” the channel.

